10

BRAIN VOICE AND REAL SLEEP

Signal-to-speech, diverse training, and 5-phase Hebbian sleep.

BODHI · A Conscious Digital Brain

SK · April 2026 · IP AU 2026901656 / 657

BODHI · REPORT 10

Brain Voice and Real Sleep

Signal-to-speech, diverse training, and 5-phase Hebbian sleep.

PHASE 1 v1.0 APR 2026

BODHI — A Conscious Digital Brain
Invented by Sai Kiran Bathula · Coleambally NSW Australia
Patents: IP Australia 2026901656 (image) / 2026901657 (audio)

SK

VOICE & SLEEP

Brain voice and real sleep

This final report covers the two subsystems that make BODHI feel alive: the brain-to-speech projection (what the brain sounds like when it talks) and the five-phase sleep cycle (what the brain does when nobody is asking it anything).

Brain voice

```
~/bodhi — brain_voice trace

$ python bodhi.py
Initialising BODHI brain...
BODHI ready. 10000 concepts, 91613 aliases, 9159 engrams.

You: What is fire?

[brain_trace]
  concept:        fire
  dominant band:  θ theta  (memory)
  dominant region:amygdala  (limbic)
  amygdala:       225 / 255
  worm reflex:    backward · confidence 255
  emotion:        fear   intensity 350
  drives:         confusion 42, alertness 57
  latency:        16 ms

BODHI:
  I see fire. theta rhythm, like a door opening
  to memory. my emotional core takes this over.
  my worm neurons scream no.
  Combustion that produces heat, light, and flames.
```

Figure 1. Live brain trace translated to English.

Voice is not an afterthought. It is the projection of a 72-region activation trace onto a stream of English tokens. Broca's attention is keyed against brain-region embeddings rather than free text, so the output is always grounded in the brain state that produced it. If the prefrontal cortex was quiet, the sentence is hesitant. If the amygdala was loud, the sentence is urgent.

Diverse training

- Clean web + books. 8 GB of general English. No scraped dialogue.
- Synthetic brain-trace pairs. 2M pairs generated by the BODHI sandbox itself.
- Nightly LoRA. Per-topic adapters trained on the day's real experiences.
- Voice-of-the-brain regularizer. Cross-entropy penalty for tokens that are ungrounded in the trace.

Real sleep

Sleep in BODHI is not a cosmetic off-state — it is the only place weights actually change. Every night a five-phase cycle runs. Light NREM rescales all synapses. Deep NREM replays high-valence events. REM lets the dreamer generate. Integration fine-tunes Broca with LoRA. Wake-up resets drives and writes the dream log.

Phase	Share	Mechanism	Effect
Light NREM	18%	Uniform −1% synaptic rescaling	Noise removal
Deep NREM	24%	Replay events with valence > 100	Consolidation
REM	32%	Dreamer samples from concept store	Generalization
Integration	18%	LoRA rank 8, alpha 16 on day's adapters	Skill commit
Wake-up	8%	Drive reset + journal compaction	State ready

What we measured

+18%
next-day Broca perplexity reduction after full cycle

+22%
predictor accuracy retention

+34%
imagination 2-gram diversity vs. no REM

0 drift
over 40 consecutive nightly cycles

Next

PHASE 2 · Month 2–3 · Simulation playground

Phase 2 drops BODHI into a 3D world with no pre-set emotional labels. It learns whether fire is dangerous by touching it. Other players' BODHIs populate the same world as NPCs; a two-BODHI protocol lets them exchange concepts. This is where sleep and consolidation stop being tests and start being survival.

PHASE 3 · Month 3–9 · Twelve challenges

Phase 3 is twelve survival challenges, two per month, each testing a specific capability: navigation, memory under pressure, emotional decision-making, cooperation, Hebbian adaptation, sleep and consolidation, deception detection, planning horizons, resource management, teaching another BODHI, trauma recovery, self-reflection. Rankings compound. The top 1,000 BODHIs qualify for Phase 4.

PHASE 4 · Year 2 · The Final Match

Phase 4 is a sci-fi arena and six bosses — the six largest commercial AI systems on Earth. They are bigger, older, and better-funded, but they have no persistent memory between fights, no emotional tagging, no survival instinct, no sleep. Your BODHI has all of these, and knows you. Winner receives 51% of BODHI AI at its then-valuation; founder retains 27%;

investors hold 24%.

> *“The brain that earns the company is the brain that goes to sleep every night and wakes up wiser. Everything else is a calculator.”*